INCH-POUND

MIL-C-70729 (AR)
31 MARCH 1989

MILITARY SPECIFICATION

CHARGE PROOF, 155MM, PXR-6297A1 LOADING, ASSEMBLING AND PACKING

This Specification is approved for use within the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading, assembling and packing for one type of propelling charge designated as Charge, Proof, PXR-6297A1 for 155mm Cannon, M199/Howitzer, M198.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and Supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-223	-	Powder, Black
MIL-C-43157	-	Cloth, Spun Viscose Rayon Resin Impregnated
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-L-48176	-	Liner (Titanium Dioxide and Paraffin Wax) for Use in Cartridges

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AMD ENGINEERING CENTER (ARDEC)

12598573 - Proof Charge, PXR-6297A1, for 155MM Cannon, M199/Howitzer M198
12902708 - Packing and Marking for Container Ammo, Metal PA 103 for Charge, Proof, 155mm, PXR-6297A1 for Cannon M199/Howitzer M198

(Copies of specifications, standards, drawings, publications and other Government documents required by suppliers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precendence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.1.1 Liner. The liner shall meet the requirements as specified in Dwg. 9378133 and tested in accordance with MIL-L-48176.

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3.2 Propelling charge. The propelling charge shall comply with all requirements specified in applicable drawings and specifications.

3.3 Weights

3.3.1 Propellant weight. The propellant weight shall comply with the weight specified in the loading authorization within plus or minus 0.25 ounce.

3.3.2 Charge, propelling assembly weight. The charge propelling assembly weight shall not differ by more than 2.0 ounces from the total amount specified in the loading authorization plus average weights of the case body, rear cap and igniter assembly, liner assembly, tie cords and tapes.

3.4 Dimensional gage.

3.4.1 Maximum length. The completed charge shall freely enter the prescribed gage. The gage shall be designed to inspect the maximum length.

3.5 Moisture content of black powder. The moisture content of the black powder at the igniter loading station at the time of loading shall not exceed 0.7%. The moisture content of the black powder in the basepad of the propelling charge shall be determined in accordance with 4.5.5.

3.6 Destructive pull test. The force of the adhesive bond between rear cap and igniter container shall be 200 lbs minimum when determined in accordance with 4.5.6.

3.7 Proving ground. Twenty propelling charges will be fired at 70°F for information.

3.7.1 Residue. The propelling charge shall function satisfactorily without excessive or smoldering residue (see 6.9).

3.8 Assembly torque. The test screw shall be assembled to withstand subsequent re-application of the minimum assembly torque specified on the applicable drawing, without moving (see 6.12).

3.9. Workmanship. All parts shall be fabricated and assembled in a thorough, workmanlike manner. They shall be free of dents, cracks, excessive adhesives not cleaned from joints and other foreign material.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absense of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality conformance inspection.

4.2.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provision of MIL-A-48078. In addition, inspection lots of propelling charge assemblies shall contain:

a. Propellant from not more than one lot number.

b. Black powder from not more than two lot numbers with the same interfix number from one manufacturer. (See 6.13).

c. CBI powder from not more than two lot numbers with the same interfix number from one manufacturer. (See 6.13).

d. Cases from not more than one interfix number from one manufacturer.

4.2.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Continuous sampling in accordance with MIL-STD-1235 may be used if approved by the Government.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70729 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9370161
4.2.2.1	End, Base Igniter				NEXT HIGHER ASSEMBLY 9370162
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cloth improper (prior to cutting of basepad)	1/		3.1	4.5.7
102	Diameter of base igniter end		0.40%	3.2	Template
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

1/ See paragraph 4.5.8

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70729 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9370162
4.2.2.2	Base Igniter Assembly				NEXT HIGHER ASSEMBLY 9370163
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of inner seam circle		0.40%	3.2	Template
102	Diameter of outer seam circle		0.40%	3.2	Template
103	Inner cavity quilt missing		0.40%	3.2	Visual
Minor					
201	Tapes not equally spaced (4 places)		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70729 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9370163
4.2.2.3	Base Igniter Loading Assembly				NEXT HIGHER ASSEMBLY 9345092
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Assembly incorrect or damaged to the extent that CBI or black powder can escape (including any seams or openings incompletely stitched or broken threads)		100%	3.2	Visual/Manual
Major					
101	Weight of black powder		100%	3.2	4.5.1
102	Weight of CBI		100%	3.2	4.5.2
103	Any seam improper		0.40%	3.2	Visual
104	Moisture content of black powder	1/		3.5	4.5.5
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ See paragraph 4.4.3.3

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70729 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.2.2.4	Rear Cap and Igniter Assembly				9345092
					NEXT HIGHER ASSEMBLY 12598573
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Damaged to the extent that black powder or CBI can escape		100%	3.2	Visual
Major 101	Damaged rear cap (cracks that affect the structural strength and dents that affect the assembly or perforate the material).		100%	3.2	Visual
102	Damaged igniter container (cracks that affect the structural strength and dents that affect the assembly or perforate the material).		100%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70729 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9345092
4.2.2.4	CONTINUATION SHEET				NEXT HIGHER ASSEMBLY 12598573
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
103	Rear cap and base igniter loading assembly not properly assembled to the igniter container, adhesive not visible for majority of circumference, more than 1/8 in. of tape end (4 places) exposed outside the assembly		100%	3.2	Visual
104	Destructive pull test	1/		3.6	4.5.6
Minor 201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ See 4.4.3.2

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70729 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9370160
4.2.2.5	Lead Laminated Rayon Cloth				NEXT HIGHER ASSEMBLY 9345104
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Established length		0.40%	3.2	Scale
Minor 201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

AMSMC[illegible]rm 1570, 1 Feb 85 Repl[illegible]s DRSMC-QA (D) Form 160, 1 Aug 83, which may not [illegible]used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70729 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12910074
4.2.2.6	Propellant and Liner Assembly				NEXT HIGHER ASSEMBLY 12598573
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Propellant not properly bundled and not tied tightly (2 places of lacing cords and 1 place of the tape prior to liner assembly assembled)		0.40%	3.2	Visual
102	Propellant weight (prior to assembly of liner) Note 1		100%	3.3.1	See 4.5.3
103	Liner assembly improper, improperly assembled, missing or improperly located, lead surface not facing propellant and middle lacing cord loose or not abutted against end of liner		100%	3.2	Visual/Scale
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ One stick of propellant may be allowed to break up to achieve the propellant weight tolerance and the broken stick shall be put in the middle of the propellant bundle.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.2.2.7	Proof, Charge, PXR-6297A1 for 155MM Cannon M199/Howitzer M198	1 OF 2	12598573
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Base igniter loading assembly missing, damaged or improper to the extent that black powder or CBI can escape (to be inspected from inside of the rear cap and igniter assembly and before assembled to the body).		1/	3.2	Visual-Manual
2	Base igniter loading assembly damaged or improper to the extent that black powder or CBI can escape (to be inspected from outside of assembly and after assembled to the body)		100%	3.2	Visual - Manual
Special	Proving ground				
a	Pressure	20		3.7	4.5.8
Major					
101	Damaged case (cracks that affect the structural strength and dents that affect the assembly or perforate the material).		100%	3.2	Visual
102	Rear cap and Igniter Assembly not properly snapped in, bosses not completely inside the holes.		100%	3.2	Visual

NOTES:
1/ 100% inspection shall be done when the rear cap and igniter assembly is preassembled at a different line.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.2.2.7	CONTINUATION SHEET				12598573
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
103	Frayed buttons (after assembly button damaged to the extent that button becomes soft and cotton-like materials are sticking out) See 6.17		100%	3.2	Visual
104	End of bundle with liner not inserted first (prior to assembly of rear cap and igniter assembly)		100%	3.2	Visual
105	Dimensional gage failure		100%	3.4.1	Gage
106	Propelling charge assembly weight		100%	3.3.2	4.5.4
Minor					
201	Markings missing, misleading or unidentifiable		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.9	Visual
Informational					
301	Moisture content of Black Powder	1/		3.5	4.5.5

NOTES:

1/ One basepad per eight (8) hour shifts

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70729 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9362602/9362603
4.2.2.8	Strap and Filler Assembly				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of filler (Dwg 9362603-1)		0.40%	3.2	Gage
Minor					
201	Strap not centrally located (Dwg 9362603)		0.65%	3.2	Visual
202	Chain-off length incorrect (Dwg 9362603) if applicable		0.65%	3.2	Scale
203	Minimum strap length (Dwg 9362602)		0.65%	3.2	Scale
204	Strap width (Dwg 93620602)	1/	0.65%	3.2	Scale
205	Marking missing, misleading or unidentifiable (Dwg 9362603)		0.65%	3.2	Visual
206	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ If a slitting machine is utilized, in lieu of the specified AQL, during the slitting process each roll will be inspected within the first 10 feet and the last 10 feet; i.e., two measurements per roll.

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.2.2.9	Container, Prior to Insertion of Charge				12902708
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Foreign matter or rust in container		100%	3.2	Visual
102	Bottom cushion missing or improper (type or thickness)		100%	3.2	Visual
103	Damaged container (to include dents protruding greater than 1/4" into container)		100%	3.2	Visual-Gage (see 6.4)
Minor	None				

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70729 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.2.2.10	Container , Prior to Sealing		12902708
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Warning label (part no. 12912974) missing		100%	3.2	Visual
Major					
101	Any component missing or improper		0.40%	3.2	Visual
102	Dimension "X", minimum (in lieu of measuring Dimension "X", the measurement of Dimension "Y", maximum may be taken)		100%	3.2	Scale
103	Filler or strap missing, filler incorrect type, filler on the top of strap, or wrapper above the top of charge		100%	3.2	Visual
104	Top cushion missing or incorrect type		100%	3.2	Visual-Scale
105	Top spacer (if applicable) incorrect type		100%	3.2	Visual
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.2.2.11	Container, Sealed				12902708
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cover incompletely engaged		0.40%	3.2	Visual
102	Test screw missing		100%	3.2	Visual
103	DOD symbol missing, misleading or unidentifiable (if applicable)		0.40%	3.2	Visual
104	Test screw assembly torque		0.40%	3.2	Test (see 6.12)
105	Air pressure test		100%	3.2	4.5.10
Minor					
201	Metallic seal missing or incorrectly applied		0.65%	3.2	Visual
202	Marking missing, misleading, or unidentifiable		0.65%	3.2	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.2.2.12	Containerized Charge				12902708
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Note (a)	Tear down inspection	(b)		(c)	4.5.9
NOTES:	(a) Category is critical or major dependent upon the category of the defect inspected. (b) See paragraph 4.4.3.1 (c) Requirement paragraph is dependent upon inspection performed.				

NOTES:

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4.4.3 Testing.

4.4.3.1 Teardown inspection. Five (5) completely assembled charges shall be randomly selected from the first three consecutive lots and tested in accordance with 4.5.10.

4.4.3.2 Destructive pull test. Five (5) rear cap and igniter assemblies (Dwg 9345092) shall be the same charges used from the teardown inspection (see 4.4.3.1) and tested in accordance with 4.5.6.

4.4.3.3 Moisture content of black powder. At least one sample of black powder from each eight hours loading of black powder spot assemblies will be tested for moisture content.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.3 of this specification.

4.5 Methods of inspection.

4.5.1 Weight of black powder. The inspections below are to be conducted by inspection personnel using equipment designated for inspection purposes and shall be clearly distinguished from production operations. The weight of the black powder will be inspected 100 percent. The weight of the black powder in the spot will be determined by subtracting an average weight of the empty assembly (no CBI or black powder) or the actual weight as marked on the individual assembly from the loaded assembly weight.

4.5.2 Weight of CBI. The inspections below are to be conducted by inspection personnel using equipment designated for inspection purposes and shall be clearly distinguished from production operations. The weight of the CBI will be inspected 100 percent. The weight of the CBI in the basepad will be determined by subtracting an average weight of the empty assembly (no CBI) or the actual weight as marked on the individual assembly from the loaded assembly weight.

4.5.3 Propellant weight. The inspections are to be conducted by inspection personnel using equipment designated for inspection purposes and shall be clearly distinguished from production operations. Any propelling charge which fails to comply with the applicable weight requirement (see 3.3) shall be classed defective and

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removed from the lot. Inspection equipment shall be in accordance with 4.4.4. The propellant bundle (without liner assembly) will be inspected 100 percent. The weight of the propellant will be determined by subtracting an average weight of the tapes from the propellant bundle (without liner assembly).

4.5.4 Charge, propelling, assembly weight. The average weights of the case body, rear cap and igniter assembly, liner assembly, cords and tapes, together with the nominal propellant weight as specified in the loading authorization shall be subtracted from the assembled charge, propelling, assembly weight. If the resulting weight deviation is greater than the requirement, the propelling charge will be classified defective and removed from the lot (see 6.11).

4.5.5 Moisture content of black powder. The moisture content of the black powder will be determined in accordance with MIL-P-223. If any sample fails to meet the specified requirement, the eight hours production represented by the sample shall be rejected. Subsequent samples taken from basepad at time of charge assembly will be for informational purpose.

4.5.6 Destructive pull test. This test shall be performed after the adhesive bond between the igniter container and rear cap has had sufficient time to set. The assembly shall be held in a suitable test fixture clamping on the igniter container. The test apparatus shall apply a longitudinal load distributed over 360° around the lip of the rear cap. Observation shall be for the force required for separation of the rear cap from the igniter container. If any sample fails to meet the specified requirement, the lot shall be rejected.

4.5.7 Cloth, spun viscose rayon resin impregnated. Each roll, as a minimum, shall be tested for compliance with the physical characteristics listed in Table I in 3.5 of MIL-C-43157. Testing will be in accordance with 4.3.1 of MIL-C-43157. Any sample which fails to comply with the applicable requirement shall be cause for rejection of the cloth bolt represented by the sample. The method of inspection to be used during production is to be established by the contractor and approved by the government. The method, as a minimum, shall contain positive assurance that the material being utilized has been verified for compliance with the specification requirements as stated above (see 6.14).

4.5.8 Proving ground assessments. These tests shall be conducted at a government proving ground in accordance with the applicable acceptance test procedure using M549, RA projectiles

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inert loaded within 0.1 pound of an as fired weight of 96.0 pounds or M864 projectiles (103.0 lbs) using primer, M82 and fired from M198 Howitzer with cannon M199 with 95 percent (minimum) remaining life. Firing shall be conducted with a spindle to charge base stand-off of 1 inch (plus or minus 1/4 inch). Prior to firing, all charges shall be temperature conditioned in a large conditioning box with adequate air circulation at the required temperature for a minimum of 24 hours and assurance will be made that, when fired, the charges are at the conditioning temperature. The following tolerances shall apply 70°F conditioning: 70°F from 68°F to 72°F. Copper crusher gages (2) shall be used to measure the maximum chamber pressure of each charge. Velocity coils will be used to measure the muzzle velocity of each charge.

4.5.8.1 Pressure. Two pressure-time gages at different locations (one in the spindle and one at the projectile base) will be used to record pressure-time traces. Differential pressure traces will be obtained by subtracting the projectile gage output from the spindle gage output. The maximum differential pressure of any charge will be defined as the maximum amplitude of the first negative wave of the differential pressure trace prior to peak pressure. Any trace which contains no amplitude in the negative direction will be considered to have a differential pressure of zero. Ignition delay will be recorded on all differential pressure firings. Twenty (20) charges will be conditioned to 70°F and ten (10) charges will be fired with the M549 (inert loaded) and ten (10) with the M864 (inert). Ten (10) calibration charges will be fired intermixed (five (5) with each projectile) with the test samples and the muzzle velocity and pressure of all test charges correctd to standard conditions.

4.5.8.2 Residue (see 6.9). All propelling charges fired at different temperatures in accordance with 4.5.8.1 shall be observed for excessive residue and smoldering residue. A test lot shall be referred to the procuring activities with a copy of test data forwarded to the technical agency for disposition whenever any of the following conditions exist:

a. Flaming, smoking or smoldering residue in a stationary position on the breech surface, chamber, or tube surface after the firing pulse.

b. Evidence of two or more rounds exhibiting excessive residue. (see 6.9.1)

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4.5.9 Teardown inspection. Each container and propelling charge will be subjected, to each of the critical (except critical defect 3, 4.2.2.7) and 100 percent major (except major defect 101, major defect 102, 4.2.2.3; major defect 102, major defect 103, 4.2.2.6; major defect 107, major defect 108, 4.2.2.7; major defect 102, 4.2.2.10) inspections defined in 4.2.2 of this specification. Samples may be accumulated or inspected upon selection providing the selection remains random and represents the lot. The inspections will be performed by the contractor and witnessed by the Government. The detection of any one of the above defects shall be cause for rejection of the lot represented by the sample.

4.5.10 Air pressure test. Each container shall be tested as specified on the appropriate drawing. A pressure drop equivalent to or greater than the applicable requirement shall be cause for rejection and removal from the lot.

5. PACKAGING

5.1 Packing

5.1.1 Level A. The propelling charges shall be packed, sealed and tested in accordance with DWG 12902708.

5.1.2 Level B. Same as specified in 5.1.1.

5.1.3 Level C. Same as specified in 5.1.1.

5.2 Marking

5.2.1 Level A, B, & C. Mark in accordance with DWG. 12902708.

6. NOTES

6.1 Intended use. The proof charges covered by this specification are intended for use with 155MM projectiles, development and acceptance tests, and for M199 cannon proof tests. This proof charge is for Proving Ground use only, to be fired in a M199 cannon with all personnel behind a protective barrier.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of designs for approval. See 6.2.3 of MIL-A-48078. Submit equipment designs, as required, to Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAR-I (D), Picatinny Arsenal, New Jersey 07806-5000. This address will be specified on the Contract Data

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Requirements List DD Form 1423 in the Contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination the characteristic may be measured or gaged to determine acceptability.

6.5 Inspection lot size. It is noted that the size of inspection lots of components, assemblies, or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.6 Proving ground. A copy of all proving ground data should be submitted to Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAR-R (D), AMCPM-CAWS-A, and SMCAR-AEE-BP, Picatinny Arsenal, NJ 07806-5000.

6.7 Charge manufacture. All test and calibration charges shall be manufactured in accordance with this specification.

6.7.1 Test charge. The test charges shall be loaded by the proving ground and the proving ground will furnish a complete set of operating procedures (including inspections to be performed and gage drawings) for approval to the technical agency prior to any manufacturing of test charges.

6.7.2 Calibration lot. All calibration lots shall meet the lot formation criteria of 4.2.1 of this specification except no more than one lot of CBI or BP shall be used. Because of the non-availability of long term storage containers (metal) for stick propellant and combustible case components, and in order to maintain homogeneity, the calibration lot shall be loaded by the contractor responsible for LAP of this charge in production. The entire lot will be loaded and sealed in PA 103 containers at the same time (no further proving ground testing is required to accept the calibration lot). However, charges needed for the establishment testing of calibration lot may be loaded at the proving ground.

6.8 Data cards. A copy of all data cards should be submitted to Cdr, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAR-R (D), and SMCAR-AEE-BP, Picatinny Arsenal, New Jersey 07806-5000.

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6.9 Residue.

6.9.1 Excessive residue. Excessive residue shall be defined as residue large enough to prevent chambering or otherwise hinder functioning of a subsequent round. In addition, any single piece of residue larger than 1 inch by 1 inch should be considered excessive. Any collected residue and photographs shall be forwarded to Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-AEE-BP for evaluation.

6.10 Teardown inspection. A list of all defects found during teardown inspection shall be submittted to Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAR-R (D), Picatinny Arsenal, New Jersey, 07806-5000.

6.11 Charge, propelling, assembly weight.

Weights to Be Determined	Assembly Drawing Number
Average weight of body	9345098
Average weight of rear cap and igniter assembly	9345092
Average weight of liner assembly	9378133
Average weight of the tapes and cords	9345104
Nominal propellant weight	Loading Authorization
Component weight (sum of above weights)	

The propelling charge assembly weight minus the component weight equals the "weight deviation".

6.12 Assembly torque. The assembly torque of the test screw will apply at time of loading only due to the relaxation of the gasket material.

6.13 Lot formation. Allowance for two black powder or CBI lots is for the purpose to facilitate LAP when a black powder or CBI lot is depleted and a new lot is introduced.

6.14 Method of inspection submission. Methods of inspection for resin impregnated spun viscose rayon cloth shall be submitted to Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAR-R(D), Picatinny Arsenal, New Jersey 07806-5000 for approval.

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6.15 Automated inspections or processes. When components or assemblies are inspected or produced by automated methods or punches or slitting machines where variations are known to be minimal, the contractor may submit a description of the process and process controls along with proposed new or reduced sampling for evaluation and approval to Cdr, ARDEC, AMSMC-QAR-R(D), Picatinny Arsenal, NJ 07806-5000.

6.16 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, ARRADCOM or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.17 Rework on frayed button. When the propelling charge is rejected because of frayed button, rework may be performed on the frayed button for re-accepting the propelling charge. The contractor should submit a description of the rework procedure to QA COR for evaluation and approval and with a copy furnished to Commander, AMSMC-QAR-R (D), Picatinny Arsenal, NJ 07806-5000.

6.18 Subject term (key word) listing.

Artillery
Cannon
Proof Charge
Howitzer
Projectiles

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1320-A982)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-70729 (AR)	CHARGE PROOF, 155MM, PXR-6297A1 LOADING, ASSEMBLINE AND PACKING

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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